Exercise Sheet 2: Multiple Quantum Systems

Use the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ for two qubits, and write $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. In Questions 1 and 2, select all correct answers and briefly justify them.

Question 1. Consider the two-qubit state

$$\rho_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|).$$

Which of the following describe ρ_{AB}?

(A) Pure.
(B) Separable.
(C) Mixed.
(D) Entangled.

Question 2. Consider the two-qubit state

$$\rho_{AB} = \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|).$$

Which of the following describe ρ_{AB}?

(A) Pure.
(B) Entangled.
(C) Mixed.
(D) A purification of $\frac{1}{2}I_A$.

Question 3. Let $\tau_{AB} = |\psi\rangle\langle\psi|$, where

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|01\rangle - |00\rangle + |11\rangle).$$

Which operator is $\tau_A = \mathrm{tr}_B[\tau_{AB}]$?

$$A = |1\rangle\langle 1|, \qquad B = \frac{1}{3}(2|0\rangle\langle 0| + |1\rangle\langle 1| + |0\rangle\langle 1| + |1\rangle\langle 0|),$$

$$C = |1\rangle\langle 1| + |0\rangle\langle 1| + |1\rangle\langle 0|, \quad D = \frac{1}{3}(|0\rangle\langle 0| + 2|1\rangle\langle 1| - |0\rangle\langle 1| - |1\rangle\langle 0|).$$

Question 4. Let

$$|u\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \qquad |v\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Expand $|u\rangle \otimes |v\rangle$ in the computational basis and write its column vector.
(b) Write the matrices $X \otimes I_2$ and $I_2 \otimes X$. Compute their action on $|u\rangle \otimes |v\rangle$, and identify which qubit each operator changes.

Question 5. Consider the pure two-qubit state

$$|\chi\rangle = \frac{1}{\sqrt{8}}(\sqrt{3}|00\rangle + \sqrt{3}|01\rangle + |10\rangle - |11\rangle).$$

(a) Find a Schmidt decomposition and the Schmidt rank of $|\chi\rangle$.
(b) Compute both reduced states and their eigenvalues.
(c) Is $|\chi\rangle$ a product state, entangled, or maximally entangled? Explain using the Schmidt coefficients.

Question 6. Let A be a three-dimensional system with state

$$\rho_A = \begin{pmatrix} 2/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a) Construct a purification $|\phi_{AR}\rangle$ using a qubit reference system R, and verify that tracing out R gives ρ_A.
(b) Replace the reference basis vectors $|0\rangle, |1\rangle$ in your construction by $|+\rangle, |-\rangle$. Verify that this gives a different purification of the same state.
(c) What is the smallest possible dimension of R? Explain why the zero eigenvalue of ρ_A does not require another reference basis vector.

Question 7. Alice and Bob share $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Alice measures in the X basis $\{|+\rangle, |-\rangle\}$. Bob chooses either the Z basis $\{|0\rangle, |1\rangle\}$ or the X basis.

(a) Find the joint outcome distribution for each of Bob's two choices.
(b) Find Alice's marginal distribution in both cases.
(c) Compute Alice's conditional probability of + given that Bob obtains 0 in the Z basis, and given that he obtains + in the X basis. Explain why Bob cannot communicate his choice of basis through these correlations alone.

Question 8. Consider the four Bell vectors

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

(a) Check that they form an orthonormal basis of the two-qubit space.
(b) Compute their reduced states. Can Alice distinguish which Bell state was prepared by measuring her qubit alone?
(c) Starting from $|\Phi^+\rangle$, apply $I_2 \otimes X$, $I_2 \otimes Z$, and $I_2 \otimes (XZ)$, where $Z = \mathrm{diag}(1, -1)$. Which Bell vector results in each case?

Solutions

1. **B and C: separable and mixed.** It is a convex combination of product states and has rank two (or purity $\mathrm{tr}[\rho_{AB}^2] = 1/2 < 1$).

2. **A, B, and D.** It is the rank-one projector onto $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Both reduced states are $I_2/2$; their mixedness implies that this pure joint state is entangled, and ρ_{AB} purifies $I_A/2$.

3. **B.** Tracing out Bob removes terms with different Bob indices:

$$\tau_A = \tfrac{1}{3}\big(|0\rangle\langle 0| + (|0\rangle + |1\rangle)(\langle 0| + \langle 1|)\big) = \tfrac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

4. **(a)** Expanding the tensor product gives

$$|u\rangle \otimes |v\rangle = \frac{\sqrt{3}|00\rangle + |01\rangle + i\sqrt{3}|10\rangle + i|11\rangle}{2\sqrt{2}},$$

with column vector $(\sqrt{3}, 1, i\sqrt{3}, i)^T/(2\sqrt{2})$. **(b)**

$$X \otimes I_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad I_2 \otimes X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

They give $(i\sqrt{3}, i, \sqrt{3}, 1)^T/(2\sqrt{2})$ and $(1, \sqrt{3}, i, i\sqrt{3})^T/(2\sqrt{2})$, respectively: $X \otimes I_2$ flips Alice's qubit, and $I_2 \otimes X$ flips Bob's.

5. **(a)** $|\chi\rangle = \frac{\sqrt{3}}{2}|0\rangle|+\rangle + \frac{1}{2}|1\rangle|-\rangle$ is a Schmidt decomposition, of rank two. **(b)**

$$\rho_A = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \qquad \rho_B = \tfrac{3}{4}|+\rangle\langle +| + \tfrac{1}{4}|-\rangle\langle -| = \begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}.$$

Both spectra are $(3/4, 1/4)$, the squared Schmidt coefficients. **(c)** Entangled, but not maximally: both coefficients are nonzero and unequal.

6. **(a)** $|\phi_{AR}\rangle = \sqrt{2/3}|0\rangle|0\rangle + \sqrt{1/3}|1\rangle|1\rangle$. **(b)** $|\psi_{AR}\rangle = \sqrt{2/3}|0\rangle|+\rangle + \sqrt{1/3}|1\rangle|-\rangle$. In each case, orthogonality of the reference vectors removes the cross terms, giving $\mathrm{tr}_R[|\phi\rangle\langle\phi|] = \mathrm{tr}_R[|\psi\rangle\langle\psi|] = \frac{2}{3}|0\rangle\langle 0| + \frac{1}{3}|1\rangle\langle 1| = \rho_A$. **(c)** The minimum dimension is $\mathrm{rank}\,\rho_A = 2$, by the Schmidt decomposition; the zero eigenvalue has zero amplitude.

7. **(a)** For Bob's Z basis, all four joint probabilities are $1/4$. For his X basis, $p(+,+) = p(-,-) = 1/2$ and the other two are zero, since $|\Phi^+\rangle = (|{+}{+}\rangle + |{-}{-}\rangle)/\sqrt{2}$. **(b)** Alice's marginal is $(1/2, 1/2)$ in both cases. **(c)** $p(A = + \mid B = 0, Z) = 1/2$ and $p(A = + \mid B = +, X) = 1$. Alice's unconditioned statistics are unchanged; using the conditional statistics requires Bob to communicate his outcome.

8. **(a)** Each vector has norm one; different supports or cancelling signs give zero pairwise inner products. Four orthonormal vectors form a basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$. **(b)** All have $\rho_A = \rho_B = I_2/2$, so Alice's local statistics cannot distinguish them. **(c)** The results are $|\Psi^+\rangle$, $|\Phi^-\rangle$, and $|\Psi^-\rangle$, respectively; in XZ, Z acts first.